



Modelling Financial Markets

Chapter 10: Market Microstructure



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AMM: Automated Market Maker
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CEX: Centralised EXchange
- ▣ CFTC: Commodity Futures Trading Commission
- ▣ CI: Confidence Interval
- ▣ DEX: Decentralised EXchange
- ▣ EDGE: Efficient Discrete Generalized Estimator (of the bid-ask spread)
- ▣ ETF: Exchange-Traded Fund
- ▣ FOK: Fill-Or-Kill (order)
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ GTC: Good-Till-Cancelled (order)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HFT: High-Frequency Trading
- ▣ IOC: Immediate-Or-Cancel (order)
- ▣ LLM: Large Language Model
- ▣ LOB: Limit Order Book
- ▣ NYSE: New York Stock Exchange
- ▣ OHLC: Open, High, Low, Close
- ▣ PIN: Probability of INformed trading
- ▣ SEC: Securities and Exchange Commission
- ▣ TWAP: Time-Weighted Average Price
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)

Acronyms used in this material (2/2)

- **VIX: CBOE Volatility Index**
- **VPIN: Volume-synchronised Probability of INformed trading**
- **VWAP: Volume-Weighted Average Price**
- **XRP: XRP (the native token of the XRP Ledger)**

Outline

Motivation

The Limit Order Book

The Bid-Ask Spread

Illiquidity and Price Impact

Intraday Patterns

Information and Prices

Price Impact and Optimal Execution

High-Frequency Trading and Market Stress

Crypto Market Microstructure

Liquidity at the Bucharest Stock Exchange

AI for Scientific Discovery

Summary

References

What Is Market Microstructure?

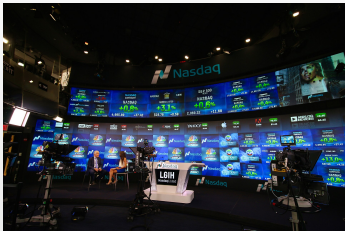
The question of this chapter

- Earlier chapters took prices as given: a daily close, a return, a volatility
- Here we ask **how** a price is formed: who trades, with which orders, under which rules
- And what trading **costs**: the spread, the price impact, the delay [Harris \(2003\)](#)

What we do in this chapter

- Describe the limit order book and simulate one
- Measure spreads and illiquidity for US stocks, BVB stocks and crypto-assets
- Explain spreads by adverse selection (Glosten–Milgrom, Kyle)
- Model price impact and optimal execution; study high-frequency trading and market stress

From the Trading Floor to the Matching Engine



Nasdaq MarketSite, Times Square, New York: a studio,
not a trading floor

Photo: Luca Marfè (2015); CC BY 2.0; Wikimedia Commons

- For two centuries prices were made by people in a room: brokers, specialists, open outcry
 - ▶ The specialist kept the order book on paper and quoted a bid and an ask
- Today almost all equity, futures and crypto trading runs on computers that match orders in microseconds
- The rules of the matching engine (priority, tick size, auctions) shape prices and costs [Gould et al. \(2013\)](#)
- Textbooks: [Hasbrouck \(2007\)](#), [Bouchaud, Bonart, Donier & Gould \(2018\)](#), [Franke, Härdle & Hafner \(2019\)](#)

Data Used in This Chapter

Series	Content	Period
SPY, 5-minute bars	ETF on the S&P 500; regular session 09:30–16:00 New York time; 1 477 full days	2020–2026
Bitcoin, 5-minute bars	24 hours a day, 7 days a week; 208 596 bars	2024–2026
US stocks, daily	SPY, Apple, Microsoft, Nvidia, JPMorgan, GameStop, Strategy	2016–2026
BVB stocks, daily	Ten blue chips and the BET ETF; values in lei converted to USD at the BNR reference rate	2016–2026
Crypto-assets, daily	Bitcoin, Ethereum, Solana, XRP, Dogecoin, Cardano, Litecoin	2016–2026
VIX, daily	Implied volatility of the S&P 500	2020–2026

- Daily data: open, high, low, close and volume; prices adjusted for dividends and splits; days without trades removed
- Order books in Section 2 and the bid-ask bounce in Section 3 are **simulated**, and labelled as such

Liquidity: Four Dimensions

- **Liquidity**: the ability to trade a quantity quickly, at a price close to the fair value
- Three dimensions named by [Kyle \(1985\)](#)
 - ▶ **Tightness**: the cost of turning a position around quickly (the bid-ask spread)
 - ▶ **Depth**: the order size needed to move the price by a given amount
 - ▶ **Resiliency**: how fast the price returns after an uninformed shock
- **Immediacy**: how long one must wait for execution [Harris \(2003\)](#)
- No single number captures all four: this chapter uses several measures and compares them

Order Types

- **Market order:** buy or sell a quantity now, at the best available prices; certain execution, uncertain price
- **Limit order:** buy at most at price p (or sell at least at p); certain price, uncertain execution
- Time conditions
 - ▶ IOC (Immediate-Or-Cancel): execute what is possible now, cancel the rest; FOK (Fill-Or-Kill): all now or nothing
 - ▶ GTC (Good-Till-Cancelled): stays in the book until executed or withdrawn
- **Stop order:** becomes a market order when the price crosses a trigger; **iceberg:** shows only part of its size
- Limit orders **supply** liquidity; market orders **consume** it

The Limit Order Book: Definitions

- **Limit order book** (LOB): the list of all standing limit orders, by price and time of arrival [Gould et al. \(2013\)](#)
- Best bid b_t (highest buy price) and best ask a_t (lowest sell price), always $b_t < a_t$
 - ▶ Mid-price $m_t = (a_t + b_t)/2$; quoted spread $a_t - b_t$; relative spread $(a_t - b_t)/m_t$, usually in basis points (1 bp = 0.01%)
- **Depth** at a price level: the total quantity waiting there
- **Tick size**: the smallest allowed price step (USD 0.01 for most US stocks priced above USD 1)
- A trade happens only when an incoming order crosses the opposite best quote

Price-Time Priority and the Matching Engine

- **Price priority:** a better price is served first
 - ▶ For buyers the highest bid, for sellers the lowest ask
- **Time priority:** at the same price, the earlier order is served first (a queue)
- Being first in the queue is valuable: a reason to invest in speed (Section 8)
- Many exchanges open and close with a **call auction**: orders are collected and matched at a single price
 - ▶ The single price maximises the executed quantity

Worked Example: A Market Order Walks the Book

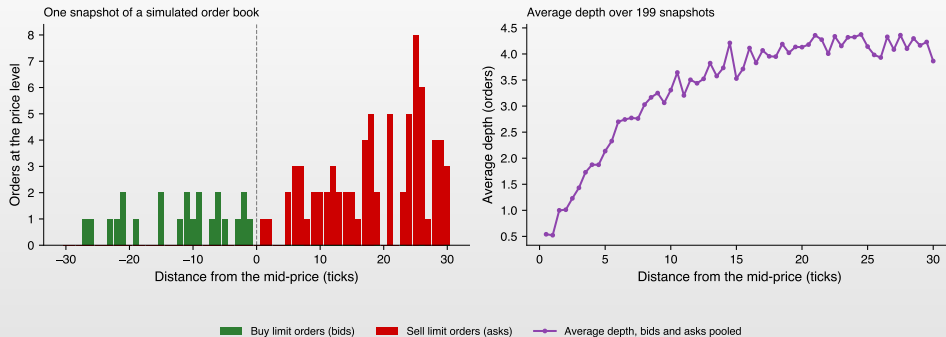
Bids		Asks	
400	99.98	100.02	300
600	99.97	100.03	500
800	99.95	100.05	400
1 500	99.90	100.08	1 000

- Mid-price $m = 100.00$; quoted spread 0.04, i.e. 4 bp
- Market buy of 800 shares: 300 at 100.02, then 500 at 100.03
 - ▶ Average price $(300 \cdot 100.02 + 500 \cdot 100.03)/800 = 100.02625$
- Cost relative to the mid-price: 2.63 bp; relative to the best ask: 0.63 bp
- After the trade the best ask is 100.05: the spread widened to 0.07

A Simple Model of the Order Book

- **Zero-intelligence** model: orders arrive at random, with no strategy [Farmer, Patelli & Zovko \(2005\)](#)
 - ▶ Limit orders: rate α per price level, placed uniformly up to 40 ticks from the opposite best quote
 - ▶ Market orders: rate μ per side; cancellations: rate δ per standing order
- Here $\alpha = 0.1$, $\mu = 0.5$, $\delta = 0.02$; 300 000 events, unit orders
- Farmer, Patelli and Zovko show that such a model already predicts much of the cross-section of spreads on the London Stock Exchange
- Use: a laboratory where spread, depth and impact can be measured exactly

A Simulated Order Book

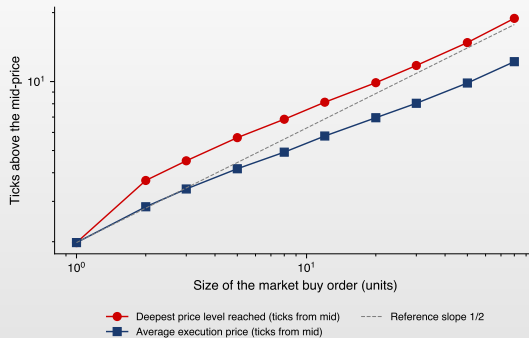


Left: one snapshot (bids green, asks red); right: average depth by distance from the mid, bids and asks pooled

Reading the Simulated Order Book

- Mean spread 3.86 ticks, median 3; in 10% of the time it exceeds 7 ticks
- Depth is thin near the mid and grows away from it
 - ▶ 0.7 orders at 1 tick, 3.3 at 10 ticks, 4.2 at 25 ticks (average over 199 snapshots)
- Reason: market orders eat the best levels first, so liquidity near the price is scarce [Tóth et al. \(2011\)](#)
- The same hump shape is observed in real order books [Bouchaud, Bonart, Donier & Gould \(2018\)](#)

The Price Impact of a Market Order



□ Market buy orders of 1 to 80 units sent against each simulated snapshot; log-log axes

Reading the Impact Function

- ▣ A 1-unit order reaches 2.0 ticks above the mid; an 80-unit order reaches 18.9 ticks
- ▣ **Concave** impact: slope 0.47 on log-log axes, close to $1/2$
 - ▶ Average execution price: slope 0.40; 80 units cost 12.2 ticks on average
- ▣ Doubling the order does not double the cost: depth grows with distance from the mid
- ▣ Real markets show a similar concave law for large orders split over time (Section 7)

Recap: The Order Book

- Limit orders supply liquidity, market orders consume it; price and time decide who trades first
- A market order pays half the spread plus the cost of walking into deeper levels
- Even random order flow creates a spread, a hump-shaped depth profile and concave impact [Farmer, Patelli & Zovko \(2005\)](#)

Why Is There a Spread?

- ▣ A liquidity provider buys at the bid and sells at the ask; the spread pays three costs [Huang & Stoll \(1997\)](#)
- ▣ **Order processing:** fees, technology, clearing
- ▣ **Inventory:** holding an unwanted position exposes the provider to price risk
- ▣ **Adverse selection:** some counterparties know more [Bagehot \(1971\)](#)
 - ▶ The provider loses to informed traders and must recover it from uninformed ones (Section 6)

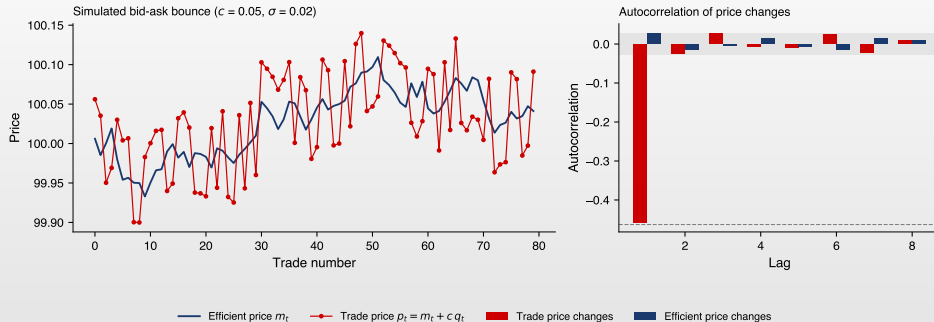
Quoted, Effective and Realised Spread

- ▣ Trade at price p_t , direction $q_t = +1$ (buyer-initiated) or -1 (seller-initiated), mid-price m_t
- ▣ **Quoted:** $S_t^Q = (a_t - b_t)/m_t$; **effective:** $S_t^E = 2q_t(p_t - m_t)/m_t$, the cost actually paid
- ▣ **Realised:** $S_t^R = 2q_t(p_t - m_{t+\Delta})/m_t$, the provider's gain after Δ (e.g. 5 minutes)
 - ▶ Price impact = $S_t^E - S_t^R = 2q_t(m_{t+\Delta} - m_t)/m_t$: the part lost to informed flow
- ▣ All three need trades, quotes and the trade direction, signed for example by the rule of [Lee & Ready \(1991\)](#)
- ▣ Without such data we **estimate** the spread from prices only: the next slides

The Roll Model

- Efficient price: random walk $m_t = m_{t-1} + \varepsilon_t$, $\varepsilon_t \sim (0, \sigma^2)$ i.i.d. [Roll \(1984\)](#)
 - ▶ Trade price $p_t = m_t + c q_t$, with half-spread c and $q_t = \pm 1$ with probability $1/2$, independent of ε
- Price change: $\Delta p_t = \varepsilon_t + c(q_t - q_{t-1})$
- $\text{Var}(\Delta p_t) = \sigma^2 + 2c^2$, $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2$
- **Roll estimator:** $\hat{s} = 2\hat{c} = 2\sqrt{-\widehat{\text{Cov}}(\Delta p_t, \Delta p_{t-1})}$
 - ▶ The bid-ask bounce creates negative autocorrelation even when the efficient price is a random walk

The Bid-Ask Bounce (Simulated)



- 5 000 simulated trades with $c = 0.05$ and $\sigma = 0.02$; grey band: $\pm 1.96/\sqrt{n}$; dashed line: the theoretical first-order autocorrelation

Reading the Roll Simulation

- ▣ True spread $2c = 0.10$; Roll estimate from the simulated trades: $\hat{s} = 0.0998$
- ▣ First-order autocorrelation -0.458 , theory $-c^2/(\sigma^2 + 2c^2) = -0.463$
- ▣ No autocorrelation at lags 2 and beyond: the bounce lasts only one trade
- ▣ The efficient price changes are uncorrelated: the market is efficient, yet trade prices are not a random walk

When the Roll Estimator Fails

- With real data the sample covariance is often **positive**: $\hat{s}$ does not exist
 - ▶ SPY, 5-minute bars: the covariance within the day is positive on 41% of days
- Causes: momentum or news trends, orders split into many trades, stale prices, bars instead of trades
- With daily data the bounce (a few bp) is buried under the daily volatility (103 bp for SPY over the last two years)
- Hence estimators based on the daily high and low: the next two slides

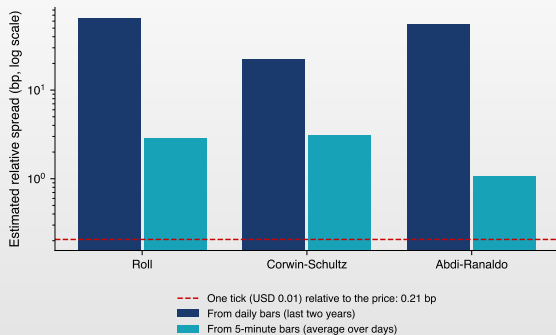
The Corwin–Schultz Estimator

- Idea: the daily high is usually a buy at the ask, the low a sell at the bid; the high-low range = volatility + spread [Corwin & Schultz \(2012\)](#)
- The volatility part grows with the length of the interval, the spread part does not: compare one day with two days
- $\beta = \sum_{j=0}^1 \left[\ln \frac{H_{t+j}}{L_{t+j}} \right]^2$, $\gamma = \left[\ln \frac{H_{t,t+1}}{L_{t,t+1}} \right]^2$, $\alpha = \frac{\sqrt{2\beta} - \sqrt{\beta}}{3 - 2\sqrt{2}} - \sqrt{\frac{\gamma}{3 - 2\sqrt{2}}}$
- $\hat{S} = \frac{2(e^\alpha - 1)}{1 + e^\alpha}$, with $H_{t,t+1}$, $L_{t,t+1}$ the high and low over the two days
 - Negative two-day estimates are set to 0, then averaged over the period

The Abdi–Ranaldo Estimator

- ▣ Uses the close c_t and the mid-range $\eta_t = (\ln H_t + \ln L_t)/2$ as a proxy for the mid-price [Abdi & Ranaldo \(2017\)](#)
- ▣ $\hat{S}^2 = 4 \mathbb{E}[(c_t - \eta_t)(c_t - \eta_{t+1})]$, with c_t the log close
- ▣ The close sits at the bid or the ask; the distance to the neighbouring mid-ranges reveals the half-spread
- ▣ If the sample mean is negative, the estimate is set to 0
- ▣ For crypto-assets, both high-low estimators track the time variation of high-frequency liquidity well [Brauneis, Mestel, Riordan & Theissen \(2021\)](#)
 - ▶ The same study finds the Amihud ratio better for **levels** and for comparing venues

Spread Estimators at Two Frequencies: SPY



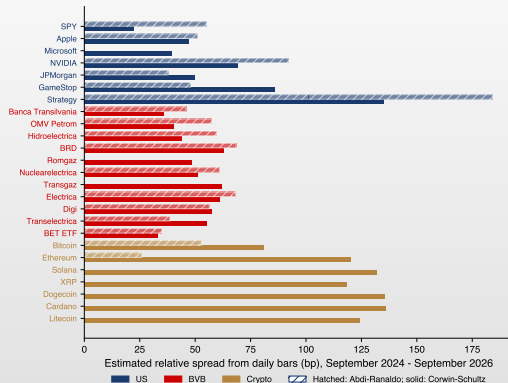
□ Daily bars: September 2024 – September 2026; 5-minute bars: estimator per day, averaged over 1 477 days (2020–2026)

 MFM_ch10_spread_estimators

Reading the Frequency Chart

- A spread cannot be smaller than one price step; with a step of USD 0.01 the bound for SPY is 0.21 bp
- From daily bars: Roll 64.3 bp, Corwin–Schultz 22.4 bp, Abdi–Ranaldo 55.1 bp
 - ▶ Two orders of magnitude above the tick: the estimators measure volatility, not the spread
- From 5-minute bars: 2.82, 3.11 and 1.06 bp: much closer, still above the tick bound
- Lesson: for very liquid assets, low-frequency spread proxies are dominated by volatility

Spread Estimates Across 25 Assets



□ Daily bars, September 2024 – September 2026; BVB stocks in lei, prices adjusted for dividends and splits

Reading the Cross-Section of Spread Estimates

- Median Corwin–Schultz estimate: US 50 bp, BVB 51 bp, crypto 124 bp
 - ▶ BVB blue chips look as liquid as US large caps: implausible, since they trade a thousand times less value (next section)
- The ranking follows volatility: crypto-assets and the most volatile stocks get the widest estimates
- Abdi–Rinaldo is 0 for 8 of the 25 assets: a negative sample mean
- Use these estimators for changes over time within one asset, not for levels across assets [Brauneis, Mestel, Riordan & Theissen \(2021\)](#)

Recap: Spreads

- The spread pays for order processing, inventory risk and adverse selection
- Bid-ask bounce makes trade prices negatively autocorrelated: the Roll estimator
- High-low estimators work at high frequency; on daily data they mostly measure volatility

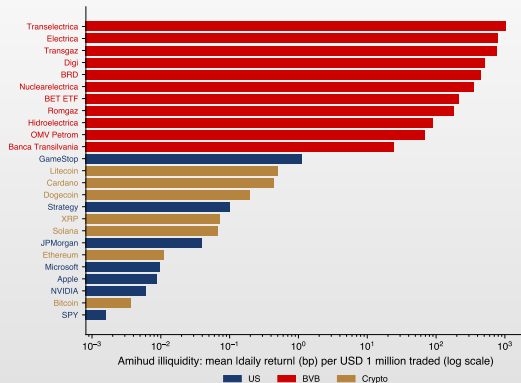
The Amihud Illiquidity Ratio

- Daily price impact per unit of money traded [Amihud \(2002\)](#)

►
$$\text{ILLIQ}_i = \frac{1}{D_i} \sum_{d=1}^{D_i} \frac{|r_{i,d}|}{\text{DVOL}_{i,d}}, \text{ with } r_{i,d} \text{ the daily return and } \text{DVOL}_{i,d} \text{ the traded value}$$

- Units here: basis points of price move per USD 1 million traded
- Needs only daily returns and volumes: available for every market and every period
- A low-frequency proxy of the Kyle price-impact coefficient (Section 6) [Hasbrouck \(2009\)](#)
- Traded value of BVB stocks: value in lei divided by the BNR reference rate of the day; crypto: traded value in USD summed across exchanges

Amihud Illiquidity Across 25 Assets

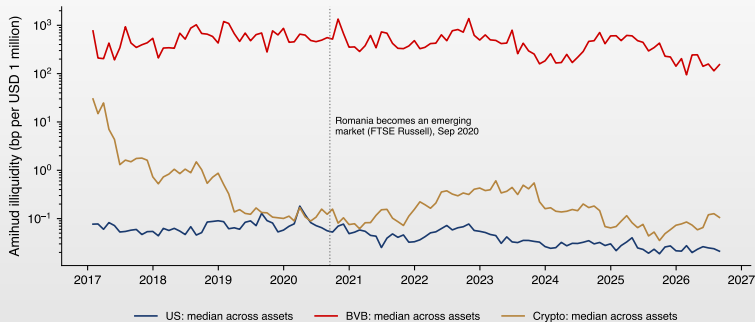


September 2024 – September 2026; log scale; each bar is the mean over about 500 trading days (730 for crypto)

Reading the Amihud Ratios

- ▣ SPY: 0.0016 bp per USD 1 million; median daily value USD 40 124 million
 - ▶ Bitcoin: 0.0037 bp, with USD 41 834 million a day across exchanges
- ▣ Banca Transilvania, the most liquid BVB stock: 24.7 bp; median value USD 3.90 million a day
 - ▶ About 15 541 times the SPY figure; Transelectrica: 1024.0 bp
- ▣ Three clusters: US large caps and the top crypto-assets, then smaller crypto-assets and GameStop, then the BVB
- ▣ Unlike the spread estimators, the Amihud ratio separates markets by orders of magnitude

Illiquidity over Time: Three Markets

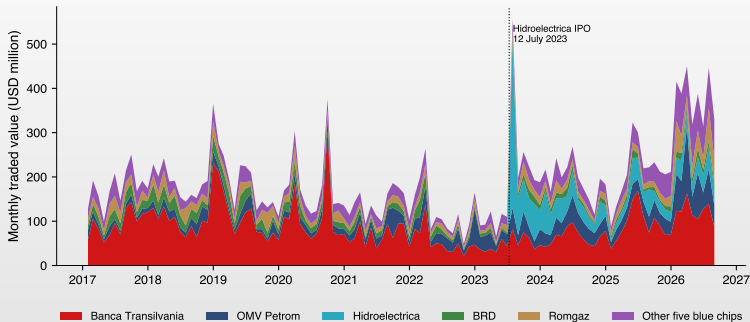


□ Monthly Amihud ratio, median across the assets of each group (without SPY and the BET ETF); log scale

Reading Illiquidity over Time

- ▣ Crypto: from 7.6 bp per USD 1 million in 2017 to 0.089 in 2026
 - ▶ Crypto trading grew from a niche to institutional volumes; spot ETFs followed (Chapter 16)
- ▣ BVB: from 435 to 169: more liquid, but still about 7 212 times the US median
- ▣ US: from 0.062 to 0.023: large caps became even cheaper to trade
- ▣ Romania moved from frontier to emerging-market status in September 2020 [BVB \(history\)](#); the BVB median shows no break at that date

Traded Value of BVB Blue Chips



□ Monthly sum of daily traded value, ten blue chips, converted to USD at the BNR reference rate

Reading the BVB Traded Value

- ▣ Ten blue chips: USD 2 095 million traded in 2017, 1 494 million in 2022, 2 520 million in 2025
- ▣ Concentration: Banca Transilvania alone made 44% of the 2025 value, Hidroelectrica 13%
 - ▶ The Hidroelectrica listing of July 2023 added a second large, liquid stock
- ▣ One day of SPY trading exceeds a whole year of the ten BVB blue chips
- ▣ Consequence for a fund: a position of a few million USD takes days to build or unwind

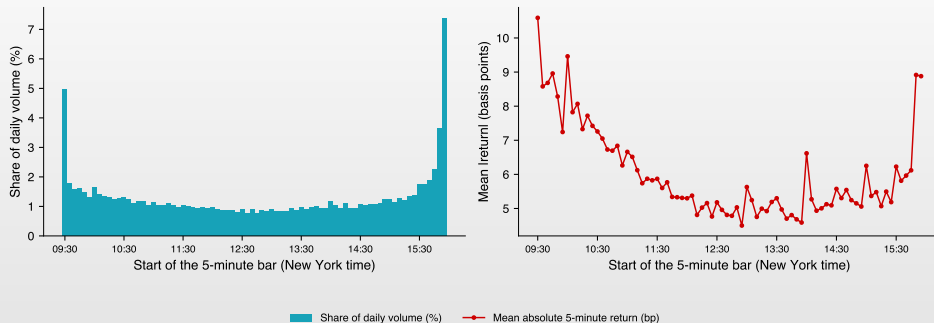
Is Liquidity Priced?

- **Level:** investors demand a higher return for assets that are costly to trade [Amihud & Mendelson \(1986\)](#)
 - ▶ Expected return increases with the spread, and concavely: long-horizon investors hold the illiquid assets
- **Time series:** expected market illiquidity raises future returns; an unexpected rise lowers current prices [Amihud \(2002\)](#)
- **Liquidity risk:** stocks that fall when market liquidity dries up earn a premium [Pástor & Stambaugh \(2003\)](#), [Acharya & Pedersen \(2005\)](#)
- Seminar Part C tests the time-series prediction on the BVB, US and crypto markets

Recap: Illiquidity

- ▣ The Amihud ratio measures price impact per unit of value traded; it separates markets by orders of magnitude
- ▣ BVB blue chips need thousands of times more price move per dollar than US large caps
- ▣ Liquidity has a level premium and a risk premium; illiquidity shocks lower prices

SPY Within the Trading Day



- 78 five-minute bars per day, 1 477 full days (October 2020 – September 2026); the first bar's return starts at the opening price

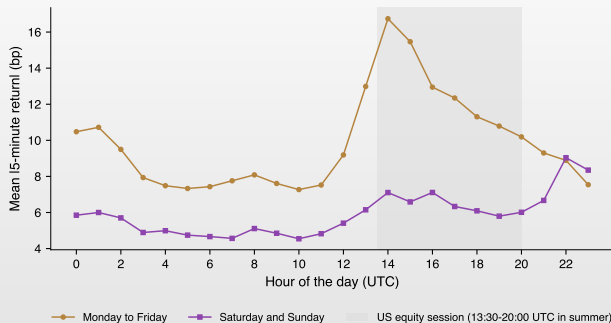
Reading the U-Shape

- ▣ Volume: the first bar carries 5.0% of the day, the last 7.4%; the last 30 minutes 18.6%
 - ▶ A midday bar (11:30–14:00) carries only 0.9% on average
- ▣ Volatility: mean absolute return 10.6 bp in the first bar, 5.1 bp at midday, 8.9 bp at the close
- ▣ The opening bar moves 2.06 times as much as a midday bar
- ▣ Spikes at 10:00 and 14:00: scheduled macro releases and Federal Reserve statements

Why a U-Shape?

- Documented for NYSE stocks by [Wood, McInish & Ord \(1985\)](#): returns and volume are highest at the open and the close
- **Open**: overnight news is absorbed; uncertainty is highest
 - ▶ Informed traders act early, before the information spreads
- **Clustering**: uninformed traders who can choose the time to trade together, and liquidity attracts liquidity [Admati & Pfleiderer \(1988\)](#)
- **Close**: index funds and ETFs trade at the closing price, which is fixed by the closing auction
- Practical rule: an execution algorithm follows the volume curve to reduce impact (Section 7)

Bitcoin Around the Clock

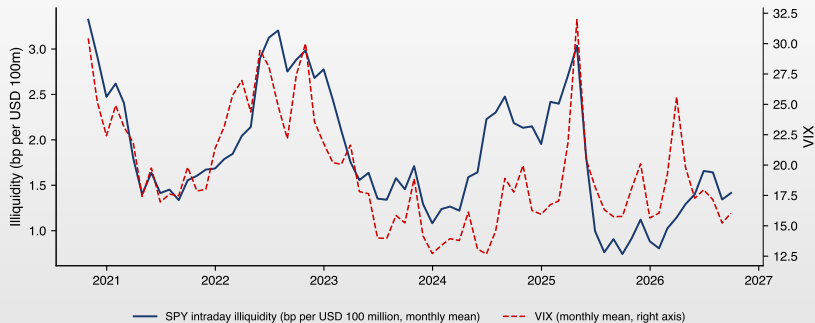


□ Mean absolute 5-minute return by hour (UTC), September 2024 – September 2026; 208 596 bars

Reading the Bitcoin Clock

- Weekdays: peak of 16.7 bp at 14:00 UTC, low of 7.3 bp at 10:00 UTC
 - ▶ The peak coincides with the opening of the US equity market and US macro releases
- Weekend: 5.9 bp against 9.9 bp on weekdays: 1.69 times calmer
- A market that never closes still follows the clock of banks, ETFs and traditional investors
- High-frequency Bitcoin data also help forecast daily risk [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)

Liquidity and Volatility: SPY and the VIX



- Daily intraday illiquidity: mean of $|r|$ / traded value over the 78 five-minute bars, in bp per USD 100 million; monthly means

Reading Liquidity and Volatility

- ▣ Correlation of daily illiquidity with the VIX: 0.58; of monthly means: 0.69
- ▣ Regression of log illiquidity on log VIX: elasticity 0.95 (HAC standard error 0.12)
- ▣ April 2025: illiquidity 3.04 against a sample mean of 1.83
 - ▶ Liquidity providers widen quotes and shrink depth exactly when investors most need to trade
- ▣ Liquidity risk is therefore correlated with market risk [Acharya & Pedersen \(2005\)](#)

Recap: Intraday Patterns

- ▣ Volume and volatility are U-shaped over the trading day, with spikes at scheduled news
- ▣ Bitcoin, open 24/7, is most active when US markets are open and calmest at weekends
- ▣ Illiquidity rises with volatility: liquidity disappears when it is most needed

Informed and Uninformed Traders

- A market maker faces two kinds of traders [Bagehot \(1971\)](#)
 - ▶ **Informed**: know something about the value; they buy when it is high and sell when it is low
 - ▶ **Uninformed** (liquidity traders): trade for their own needs, such as savings, redemptions or hedging
- The market maker cannot tell them apart, but knows that a buy order is more likely to come from an informed buyer
- So it sets the ask above the current value estimate and the bid below it: the adverse-selection spread
- Each trade reveals information: prices learn from order flow

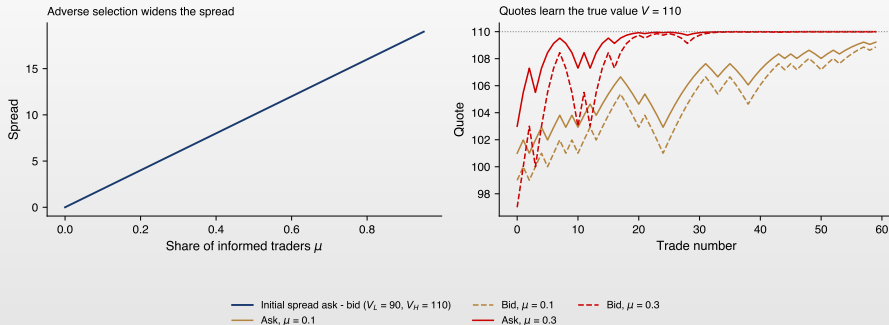
The Glosten–Milgrom Model

- Value $V \in \{V_L, V_H\}$, $P(V = V_H) = \theta$; each period one trader arrives [Glosten & Milgrom \(1985\)](#)
 - ▶ With probability μ informed (buys if $V = V_H$, sells if $V = V_L$), otherwise uninformed (buys or sells with probability $1/2$)
- Competitive, risk-neutral market maker: zero expected profit on each trade
- $a = \mathbb{E}[V \mid \text{buy}]$, $b = \mathbb{E}[V \mid \text{sell}]$
- $P(\text{buy} \mid V_H) = \mu + \frac{1-\mu}{2}$, $P(\text{buy} \mid V_L) = \frac{1-\mu}{2}$
- After each trade, Bayes' rule updates θ : a buy raises it, a sell lowers it

Glosten–Milgrom: the Spread

- $P(V_H \mid \text{buy}) = \frac{\left(\mu + \frac{1-\mu}{2}\right) \theta}{\left(\mu + \frac{1-\mu}{2}\right) \theta + \frac{1-\mu}{2}(1-\theta)}, \quad a = V_L + (V_H - V_L) P(V_H \mid \text{buy})$
- At $\theta = 1/2$: $a - b = \mu(V_H - V_L)$
 - ▶ The spread grows linearly with the share of informed traders
 - ▶ $V_L = 90$, $V_H = 110$: spread 2 for $\mu = 0.1$, 6 for $\mu = 0.3$
- No inventory, no fees: the spread exists purely because of information
- Trade prices form a martingale: the market is efficient in the semi-strong sense (Chapter 2)

Glosten–Milgrom in Action



- Left: initial spread as a function of μ ; right: one simulated sequence of 60 trades with true value $V = 110$ (same random draws for both μ)

Reading the Learning Paths

- With $\mu = 0.3$ the bid exceeds 108 after 7 trades; with $\mu = 0.1$ only after 48
- More informed traders: wider spreads, but faster price discovery
 - ▶ The trade-off between liquidity and informational efficiency
- The spread narrows as the market maker learns: less remains to be learned from the next trade
- Empirical counterpart: the probability of informed trading (PIN) [Easley, Kiefer, O'Hara & Paperman \(1996\)](#)

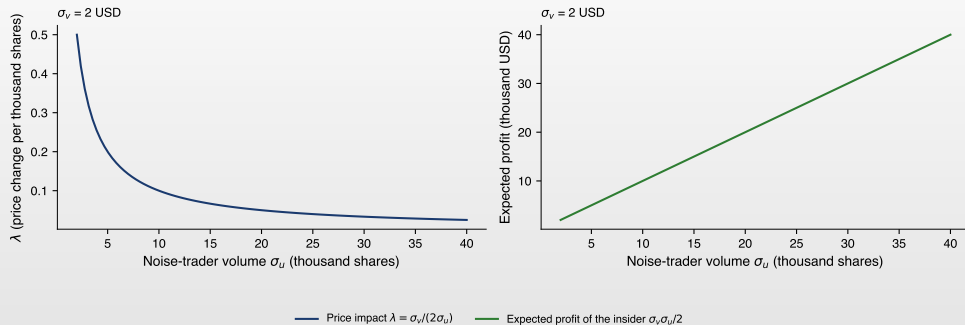
The Kyle Model

- One insider knows $v \sim N(p_0, \sigma_v^2)$ and submits a market order x [Kyle \(1985\)](#)
 - ▶ Noise traders submit $u \sim N(0, \sigma_u^2)$, independent of v
 - ▶ The market maker sees only the total order flow $y = x + u$ and sets $p = \mathbb{E}[v \mid y]$
- The insider chooses x to maximise $\mathbb{E}[(v - p)x]$, knowing that his own order moves the price
- Linear equilibrium: $x = \beta(v - p_0)$, $p = p_0 + \lambda y$

Kyle: the Equilibrium

- $\beta = \frac{\sigma_u}{\sigma_v}, \quad \lambda = \frac{\sigma_v}{2\sigma_u}, \quad \mathbb{E}[\text{profit of the insider}] = \frac{\sigma_v \sigma_u}{2}$
- λ ("Kyle's lambda"): price move per unit of order flow; $1/\lambda$ is **market depth**
 - ▶ More noise trading (σ_u up): deeper market, larger insider trades, larger insider profit
- $\text{Var}(v \mid p) = \sigma_v^2/2$: the price reveals exactly half of the private information
- The insider hides behind noise traders: his order has the same variance as their orders, $\text{Var}(x) = \sigma_u^2$

Kyle's Lambda and the Insider's Profit



□ $\sigma_v = 2$ USD; σ_u from 2 000 to 40 000 shares

From Models to Measurement

- Kyle's λ is estimated by regressing price changes on signed order flow; the Amihud ratio is its daily proxy Hasbrouck (2009)
- PIN: estimated from daily counts of buys and sells, by maximum likelihood Easley, Kiefer, O'Hara & Paperman (1996)
- VPIN: order imbalance measured in volume time, for high-frequency markets Easley, López de Prado & O'Hara (2012)
 - Rises when order flow becomes one-sided, i.e. "toxic" for liquidity providers
- Order-flow imbalance at the best quotes explains most short-horizon price changes Cont, Kukanov & Stoikov (2014)

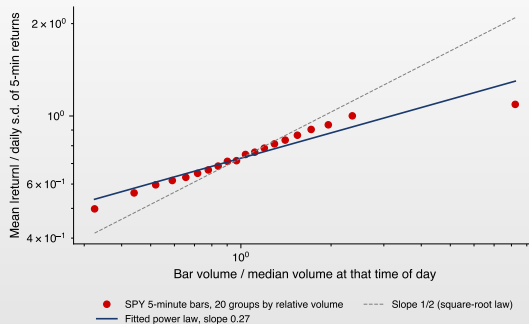
Recap: Information

- ▣ Adverse selection alone creates a spread proportional to the share of informed traders
- ▣ Prices learn from order flow; faster learning comes with wider spreads
- ▣ Kyle's $\lambda = \sigma_v / (2\sigma_u)$: depth grows with noise trading, and half of the information enters the price

Metaorders and the Square-Root Law

- ▣ A **metaorder**: a large order split into many small trades over hours or days
- ▣ Empirical law for the total price move Tóth et al. (2011), Bouchaud, Bonart, Donier & Gould (2018)
 - ▶ $\Delta P \approx Y \sigma_{\text{day}} \sqrt{Q/V_{\text{day}}}$, with Q the metaorder size, V_{day} the daily volume, Y of order 1
- ▣ Concave: trading 1% of daily volume costs about 0.1σ , not 0.01σ
- ▣ Explanation: liquidity near the price is scarce and refills slowly, as in the simulated book of Section 2
- ▣ Testing it needs the traders' own order records; public bars allow only an indirect look (next slide)

Volume and Price Moves in SPY Bars



- 115 052 five-minute bars in 20 equal-size groups by relative volume (bar volume / median volume at that time of day)

Reading the Volume–Move Relation

- ▣ Mean $|r|$, scaled by the day's volatility, rises with relative volume with a slope of 0.27 on log-log axes
- ▣ Concave, as the square-root law predicts, but flatter than $1/2$
- ▣ Not a causal test: a bar mixes buyers and sellers, and news moves both volume and price
 - ▶ Opposite trades cancel, so the net flow is smaller than the volume: this flattens the slope
- ▣ For execution costs use the metaorder law, calibrated on one's own trades

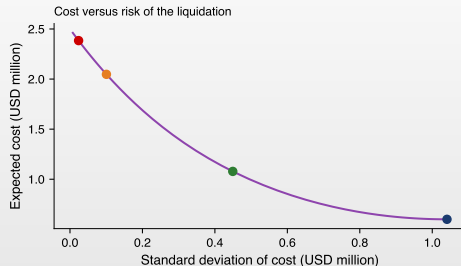
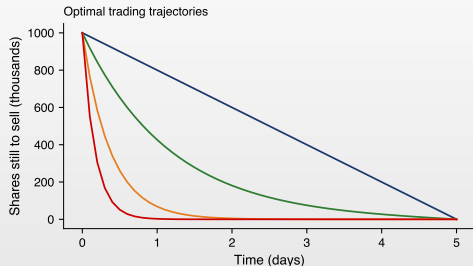
Optimal Execution: the Trade-Off

- Sell X shares within T days, in N steps of length $\tau = T/N$; holdings x_k with $x_0 = X$, $x_N = 0$; trades $n_k = x_{k-1} - x_k$
Almgren & Chriss (2001)
- Price dynamics: $S_k = S_{k-1} + \sigma\sqrt{\tau}\xi_k - \gamma n_k$; execution price $\tilde{S}_k = S_{k-1} - \eta n_k/\tau$
 - ▶ γ : permanent impact; η : temporary impact, paid only by the trade itself
- **Trading fast**: high temporary impact, low price risk; **trading slowly**: the opposite
- Objective: minimise $\mathbb{E}[C] + \lambda \text{Var}[C]$, with C the implementation cost and λ the risk aversion

Almgren–Chriss: the Optimal Trajectory

- $x_k = X \frac{\sinh(\kappa(T - t_k))}{\sinh(\kappa T)}, \quad \frac{2}{\tau^2} (\cosh(\kappa\tau) - 1) = \tilde{\kappa}^2 = \frac{\lambda\sigma^2}{\tilde{\eta}}, \quad \tilde{\eta} = \eta - \frac{\gamma\tau}{2}$
- $\mathbb{E}[C] = \frac{1}{2}\gamma X^2 + \frac{\tilde{\eta}}{\tau} \sum_{k=1}^N n_k^2, \quad \text{Var}[C] = \sigma^2\tau \sum_{k=1}^N x_k^2$
- $\lambda = 0$: $\kappa = 0$ and x_k falls linearly (trade evenly, the time-weighted schedule)
 - Large λ : most of the order is traded at the start, to cut the exposure to price risk
- $1/\kappa$ is the “half-life” of the trade: the time scale over which the position is worked off

Optimal Trajectories and the Efficient Frontier



— Risk-neutral, $\lambda = 0$ (sell evenly) — $\lambda = 2e-5$ — Efficient frontier (N = 5 days)
 — $\lambda = 2e-6$ — $\lambda = 1e-4$ (very risk averse)

- Numerical example of [Almgren & Chriss \(2001\)](#): $X = 10^6$ shares, $S_0 = 50$ USD, $T = 5$ days, $\sigma = 0.95$ USD per day^{1/2}, $\eta = 2.5 \cdot 10^{-6}$, $\gamma = 2.5 \cdot 10^{-7}$

Reading the Execution Chart

Strategy (5 daily steps)	Expected cost	Standard deviation
Even selling, $\lambda = 0$	0.60 million USD	1.04 million USD
$\lambda = 2 \cdot 10^{-6}$	1.08	0.45
$\lambda = 2 \cdot 10^{-5}$	2.05	0.10
$\lambda = 10^{-4}$	2.38	0.02

- With $\lambda = 2 \cdot 10^{-6}$ the first day sells 57% of the position; 428 599 shares remain
- Moving from even selling to $\lambda = 2 \cdot 10^{-6}$ cuts the standard deviation from 1.04 to 0.45 million USD, for 0.48 million USD more expected cost
- The frontier is convex: the first units of risk reduction are cheap, the last ones expensive

Execution Benchmarks

- ▣ **Implementation shortfall:** the difference between the paper portfolio at the decision price and the executed one [Perold \(1988\)](#)
- ▣ VWAP (Volume-Weighted Average Price): trade in proportion to the expected volume curve (the U-shape of Section 5)
- ▣ TWAP (Time-Weighted Average Price): equal amounts in each interval, the $\lambda = 0$ solution
- ▣ A participation algorithm (a fixed share of market volume) ignores price
 - ▶ This is the algorithm at the heart of the Flash Crash (next section)

Recap: Impact and Execution

- Impact is concave in size: the square-root law for metaorders
- Optimal execution trades off expected impact cost against price risk
- Risk-averse traders front-load; risk-neutral traders spread the order evenly

High-Frequency Trading



A microwave relay tower (Mojave, California); similar line-of-sight links carry trading signals between Chicago and New Jersey

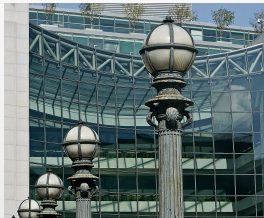
Photo: [Tony Webster \(2016\)](#); CC BY-SA 2.0; [Wikimedia Commons](#)

- HFT (high-frequency trading): computer algorithms that trade in micro- to milliseconds, end the day flat and profit from many tiny trades
- Speed: servers placed next to the exchange's matching engine (co-location), microwave links
 - ▶ Microwaves travel through air faster than light through glass fibre
- Low latency speeds up the incorporation of information and narrows spreads [Hasbrouck & Saar \(2013\)](#)

HFT Firms as the New Market Makers

- A single HFT firm acted as a market maker on Chi-X and Euronext, with most of its trades passive (limit orders) [Menkveld \(2013\)](#)
 - ▶ It earned on the spread and lost on positions: the classic market-making business, at high speed
- High-frequency markets change what “information” means: order flow and latency matter as much as fundamentals [O'Hara \(2015\)](#)
- HFT liquidity is fast to come and fast to leave: plentiful in calm markets, fragile in stress
- Market makers in limit-order markets as intermediaries: [Pele \(2025\)](#)

The Flash Crash of 6 May 2010

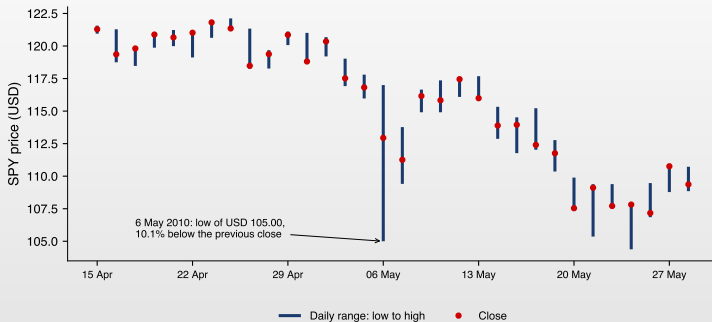


Headquarters of the US Securities and Exchange
Commission, Washington

Photo: David (dbking) (2008); CC BY 2.0; Wikimedia
[Commons](#)

- Report of the US futures (CFTC) and securities (SEC) regulators [CFTC & SEC \(2010\)](#)
- At 2:32 p.m. a mutual fund complex started selling 75 000 E-Mini S&P 500 futures (about USD 4.1 billion)
 - ▶ The algorithm targeted 9% of the previous minute's volume, "without regard to price or time"
- By 2:45:28 p.m. buy-side depth in the E-Mini had fallen below 1% of its morning level
- Over 20 000 trades in more than 300 securities were executed more than 60% away from their earlier prices, some at a penny

SPY Around the Flash Crash

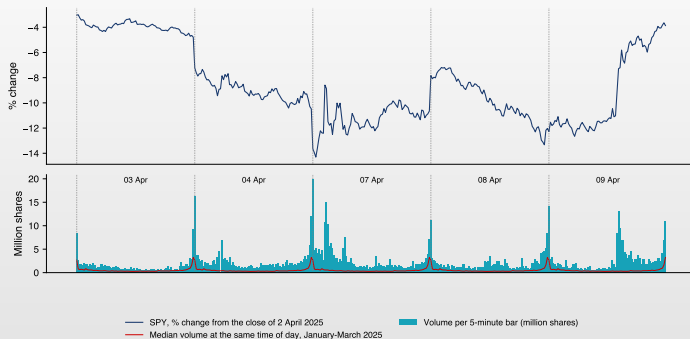


□ Daily low-high range and close, 15 April – 28 May 2010 (prices not adjusted)

Reading the Flash Crash

- ▣ SPY low of USD 105.00: 10.1% below the previous close of 116.82; it still closed only 3.3% lower
- ▣ Daily volume was 3.0 times the median of the previous three weeks
- ▣ HFT firms did not trigger the crash, but their reaction to the selling pressure amplified it [Kirilenko, Kyle, Samadi & Tuzun \(2017\)](#)
 - ▶ A 5-second pause in E-Mini trading at 2:45:28 p.m. let buyers return [CFTC & SEC \(2010\)](#)
- ▣ Responses: single-stock circuit breakers (a 5-minute pause after a 10% move in 5 minutes) and clearer rules for cancelling erroneous trades [CFTC & SEC \(2010\)](#)

A Recent Stress Episode: April 2025



- SPY, 5-minute bars, 3–9 April 2025; bottom panel: bar volume against the median volume at the same time of day in January–March 2025

Reading April 2025

- ▣ 7 April: intraday range of 8.6%; one 5-minute bar rose 4.28%, and a bar ten minutes later fell 2.92%
 - ▶ Realised volatility of the day: 6.7%, against a sample median of 0.61%
- ▣ 9 April: +10.0% from the open to the close
- ▣ Over the five days, volume was 4.6 times its usual level at the same times of day
- ▣ Unlike 2010, prices stayed continuous: circuit breakers were not needed, but liquidity was thin (Section 5)

The Speed Race and Frequent Batch Auctions

- In a continuous order book, correlations between related markets break down at millisecond horizons [Budish, Cramton & Shim \(2015\)](#)
 - ▶ This creates arbitrage opportunities that go to the fastest trader: an arms race in speed
- The cost of the race is borne by investors through wider spreads and thinner books
- Proposal: **frequent batch auctions**, e.g. every second, at a single price
- Competition moves from speed to price: narrower spreads and deeper markets in the model

Recap: HFT and Stress

- HFT firms are today's market makers: cheap liquidity in normal times, fragile under stress
- 6 May 2010: a price-insensitive sell algorithm met a book that emptied; prices fell 10% in minutes
- Market design (auctions, pauses, tick sizes) can reduce the value of pure speed

Crypto Exchanges

- ▣ Trading never stops: 24 hours a day, 7 days a week, with no opening or closing auction
- ▣ CEX (Centralised EXchange): a firm runs a limit order book, as on a stock exchange
 - ▶ The same asset trades on dozens of venues at once: liquidity is fragmented
- ▣ DEX (Decentralised EXchange): trades run by smart contracts, often without an order book: an automated market maker (AMM) prices by a formula (Chapter 16)
- ▣ The daily traded values used here sum trading across exchanges

Arbitrage Across Crypto Exchanges

- ▣ The same Bitcoin can trade at different prices on different exchanges [Makarov & Schoar \(2020\)](#)
- ▣ Price gaps are much larger **across countries** than within a country
 - ▶ Capital controls limit the movement of arbitrage capital
- ▣ The gaps open when Bitcoin appreciates strongly: demand shocks hit local markets first
- ▣ A common component of signed volume across exchanges explains 80% of Bitcoin returns [Makarov & Schoar \(2020\)](#)

Measuring Crypto Liquidity

- ▣ High-low estimators capture the **time variation** of crypto liquidity well; Amihud-type ratios capture **levels** and differences between venues [Brauneis, Mestel, Riordan & Theissen \(2021\)](#)
 - ▶ Consistent with our results: spread estimates rank crypto by volatility, Amihud by traded value
- ▣ Liquidity of crypto-assets can be forecast from market data [Agakishiev, Härdle & Pele \(2025\)](#)
- ▣ Bitcoin and Ethereum trade as much value as the largest US stocks; smaller tokens are hundreds of times less liquid
- ▣ Weekend and night hours are calmer and thinner (Section 5)

Recap: Crypto Microstructure

- Continuous, global and fragmented: many order books for one asset
- Cross-country price gaps reveal limits to arbitrage
- Top crypto-assets now rival US large caps in liquidity; the long tail does not

The Bucharest Stock Exchange



Offices of the Bucharest Stock Exchange

Photo: [Za.kurek \(2014\)](#); CC BY-SA 4.0; [Wikimedia](#)

[Commons](#)

- An electronic limit order book, as in the rest of this chapter [BVB \(history\)](#)
- Emerging-market status (FTSE Russell) since September 2020; Banca Transilvania and Nuclearelectrica were the first stocks in FTSE emerging-market indices
- A thin market: herding and bubbles in illiquid markets were studied on the BVB by [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#)
 - ▶ Bubbles and crashes are the subject of Chapter 17

BVB Liquidity: What the Data Say

- ▣ Amihud ratio of Banca Transilvania: 24.7 bp per USD 1 million; of Transelectrica: 1024.0; of SPY: 0.0016
- ▣ The BET ETF trades less than Banca Transilvania: 212.9 bp per USD 1 million
- ▣ Illiquidity has fallen since 2017, but the BVB remains thousands of times less liquid than US large caps
 - ▶ Daily spread estimates hide this: they rank BVB stocks with US stocks, because volatilities are similar
- ▣ Monthly market returns fall when illiquidity rises unexpectedly, on the BVB as in the US and in crypto (t-statistics -4.0, -3.8, -4.1; seminar Part C)

Recap: The BVB

- ▣ Concentrated trading: two stocks carry more than half of the blue-chip traded value
- ▣ Use traded-value measures (Amihud), not daily spread proxies, to compare the BVB with other markets
- ▣ For portfolio decisions, the illiquidity of the BVB is a cost and a risk (Chapter 4 and seminar Part C)

An Open Question: Do Price-Based Proxies Measure Trading Costs?

- Where trades and quotes are unavailable (BVB history, many crypto venues), costs are estimated from daily OHLC bars
 - ▶ Jahan-Parvar & Zikes (2023): low-frequency estimators track effective spreads only when the spread is large relative to volatility; otherwise volatility dominates
 - ▶ Ardia, Guidotti & Kroencke (2024): EDGE, an estimator from open, high, low and close prices, unbiased and efficient under their assumptions
- Formal question: $\hat{S}_i = S_i + b(\sigma_i) + \varepsilon_i$ across assets i
 - ▶ H_0 : $b \equiv 0$ (the proxy ranks assets by cost) against H_1 : b increasing in σ_i
 - ▶ Identification needs a cost benchmark independent of volatility: effective spreads from trades and quotes, or a price-impact measure such as Amihud
- Open: the spread-to-volatility ratio is lowest exactly where proxies are needed most (volatile, thinly traded markets)

The Discovery Loop, Step by Step (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: [Semantic Scholar](#), [Elicit](#), [Google Scholar](#)
- 1. **Literature**: collect validation studies, then check every DOI on [Crossref](#) (the title must match)
 - ▶ Find studies that validate Roll, Corwin-Schultz, Abdi-Ranaldo and EDGE against effective spreads from trades and quotes; report the benchmark, the correlation in levels and in changes, the markets, and the DOI.
- 2. **Competing hypotheses**, each with a cross-sectional prediction
 - ▶ Give rival explanations for why Corwin-Schultz ranks BVB blue chips next to US large caps, and for each the predicted Spearman correlation of the estimate with volatility, with the Amihud ratio and with traded value.
- 3. **Data and code**: replicate the chapter's numbers first
 - ▶ Write Python that reads `data/market/<symbol>.csv` for the 25 assets of this chapter and computes Corwin-Schultz exactly as in the paper, with negative two-day estimates set to zero. Target medians: US 50, BVB 51, crypto 124 bp

The Discovery Loop, Step by Step (2/2)

- 4. **Robustness**, pre-registered
 - ▶ Write a pre-registered plan: Spearman and partial Spearman correlations, bootstrap over assets, leave-one-market-out (US, BVB, crypto), a second window 2022-2024, and EDGE and Abdi-Ranaldo in place of Corwin-Schultz.
- 5. **Critique**: ask the assistant to attack the benchmark itself
 - ▶ Act as a referee: argue that the Amihud ratio is not a cost measure either (it scales with volume and with volatility); which benchmark would identify the spread, and what would it cost to obtain?
- 6. **Report**: every estimator and window, bootstrap intervals, the benchmark's limits, code and the AI log

What the Human Must Check

- **Tick-size floor:** no spread is below one tick; test: compare every estimate with $0.01/P$ (0.21 bp for SPY)
- **Volatility contamination:** test: partial correlation with the benchmark controlling for volatility, and the 5-minute versus daily comparison of Section 3
- **Truncation:** negative estimates set to zero bias the mean upwards; report the share truncated and a median
- **Units and venues:** lei converted at the BNR rate; crypto volume summed over exchanges, some with inflated volume; test: rerun on a subset of venues
- **Small, clustered cross-section:** 25 assets in 3 markets are not exchangeable; test: bootstrap and leave-one-market-out
- **Mistyped formulas:** check the constants of the estimator against the paper and run a unit test on simulated prices with a known spread

Mini-Case: Cost or Volatility?

- Replication: median Corwin–Schultz estimates US 50, BVB 51, crypto 124 bp (Section 3)
- A typical assistant proposal: “the estimates are similar, so BVB blue chips are about as liquid as US large caps”. If the proxy measures cost, its ranks should follow the Amihud ranks
- Test on the 25 assets, September 2024 – September 2026 (Spearman rank correlations)
 - ▶ With daily volatility: $\rho = 0.93$ ($p < 0.001$); with the Amihud ratio: $\rho = -0.11$ ($p = 0.59$)
 - ▶ Difference 1.04, bootstrap over assets 95% CI [0.54, 1.44]; partial ρ with Amihud given volatility: 0.24
- Interpretation: daily Corwin–Schultz ranks assets by volatility, not by cost; the liquidity claim is rejected, as [Jahan-Parvar & Zikes \(2023\)](#) predict for a low spread-to-volatility ratio
- Code: `Quantlets/Ch_10/ai_discovery_case.py`

Your Turn: A Project Seed

- Question: which estimator from daily bars, if any, measures the level of trading costs on the BVB and for crypto-assets?
- Replicate first: SPY Corwin–Schultz from daily bars 22.4 bp versus 3.11 bp from 5-minute bars and a tick floor of 0.21 bp (Section 3)
- Deliverables
 - ▶ A pre-registration (hypotheses, estimators, benchmark, tests, corrections) dated before the first run
 - ▶ EDGE implemented exactly as in [Ardia, Guidotti & Kroencke \(2024\)](#), validated on simulated prices with a known spread, then compared with the other estimators on the 25 assets
 - ▶ Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in `AI_USE.md` and every error the assistant made in `AI_ERRORS.md`:
github.com/danpele/MFM/tree/main/project

Summary

- ▣ Prices are formed in limit order books: limit orders supply liquidity, market orders consume it
- ▣ The spread pays for processing, inventory and adverse selection; Glosten–Milgrom and Kyle model the last one
- ▣ Spread proxies from daily data measure volatility; the Amihud ratio separates markets by orders of magnitude
- ▣ Liquidity follows the clock (U-shape) and the VIX; it disappears in stress (2010, 2025)
- ▣ Impact is concave in size; optimal execution balances impact against price risk

Which Liquidity Measure When?

Available data	Suggested measure
Trades and quotes with timestamps	Effective and realised spread, price impact, order-flow imbalance
Intraday bars (1–5 minutes)	Corwin–Schultz or Abdi–Ranaldo per day; intraday Amihud
Daily prices and volumes	Amihud ratio for levels across assets; high-low estimators only for changes over time
Own executions	Implementation shortfall; calibrate the square-root law
Theory, what-if analysis	Simulated order books; Glosten–Milgrom, Kyle, Almgren–Chriss

Coming Next: Continuous-Time Models (Chapter 11)

- ▣ From trades and order books back to smooth models of prices: Brownian motion and Itô calculus
- ▣ At high frequency, microstructure noise contaminates volatility estimates: realised volatility (Chapter 9) needs care
- ▣ Further reading: Hasbrouck (2007), Bouchaud, Bonart, Donier & Gould (2018), Harris (2003), Franke, Härdle & Hafner (2019)

Further Study on Quantinar

Advanced courses on quantinar.com

 Applied Computational Finance

Code and data

 Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_10

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